Machine Learning (Week 2)

End-to-End Energy Dataset Training Workflow

SourceCode:

https://github.com/MYethishwar/AI-engineering/blob/main/Week%202%20Machine%20Learning/machine_learning_done_when.ipynb

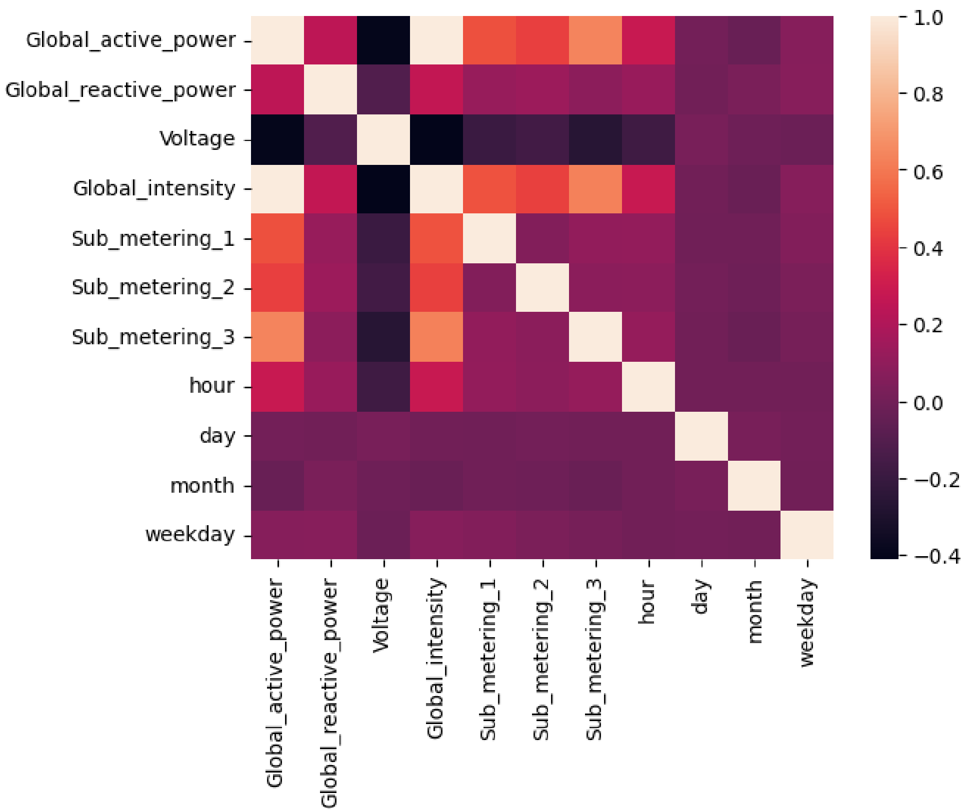
Live HuggingFace Deployment APP:

https://huggingface.co/spaces/MYethishwar/Energy-Prediction-System-Using-Linear-Logistic-Regression

Sample Dataset:

	Datetime	Global_active_power	Global_reactive_power	Voltage	Global_intensity	Sub_metering_1	Sub_metering_2	Sub_metering_3
0	2006-12-16 17:24:00	4.216	0.418	234.84	18.4	0.0	1.0	17.0
1	2006-12-16 17:25:00	5.360	0.436	233.63	23.0	0.0	1.0	16.0
2	2006-12-16 17:26:00	5.374	0.498	233.29	23.0	0.0	2.0	17.0
3	2006-12-16 17:27:00	5.388	0.502	233.74	23.0	0.0	1.0	17.0
4	2006-12-16 17:28:00	3.666	0.528	235.68	15.8	0.0	1.0	17.0

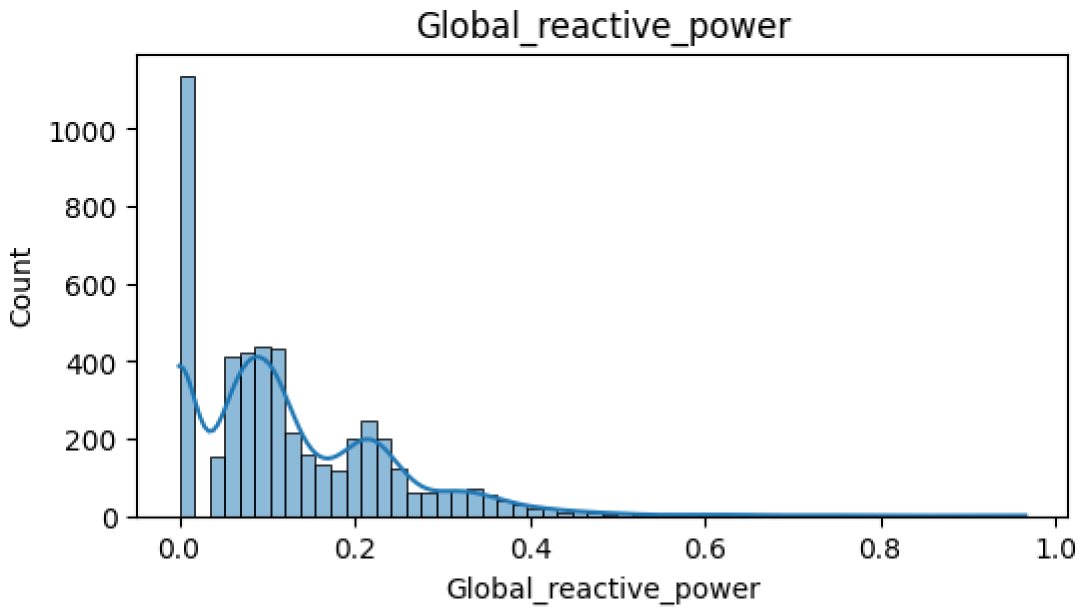
Relationship between features:

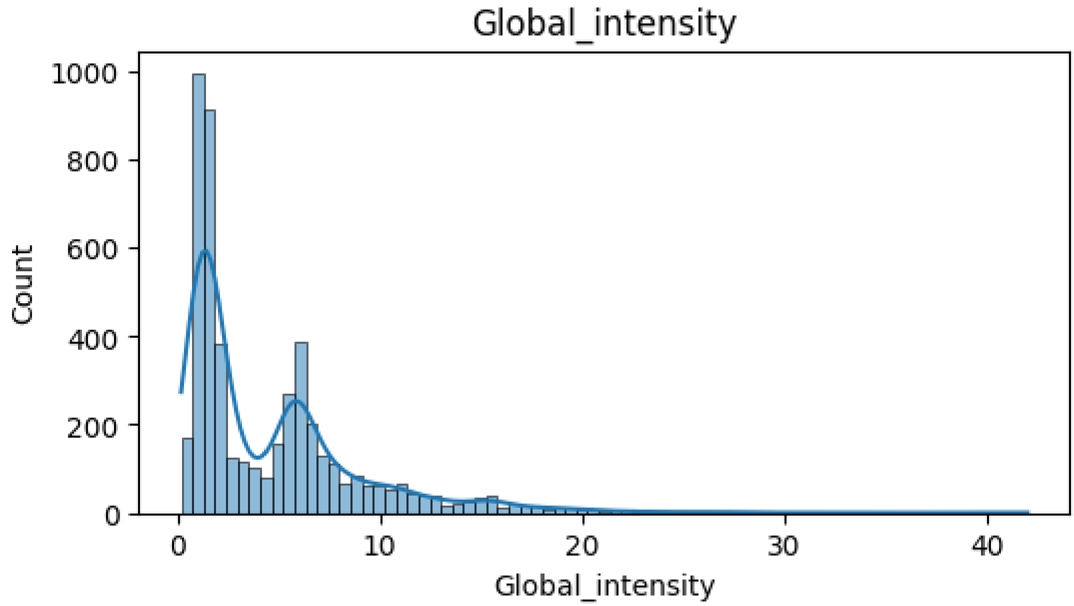


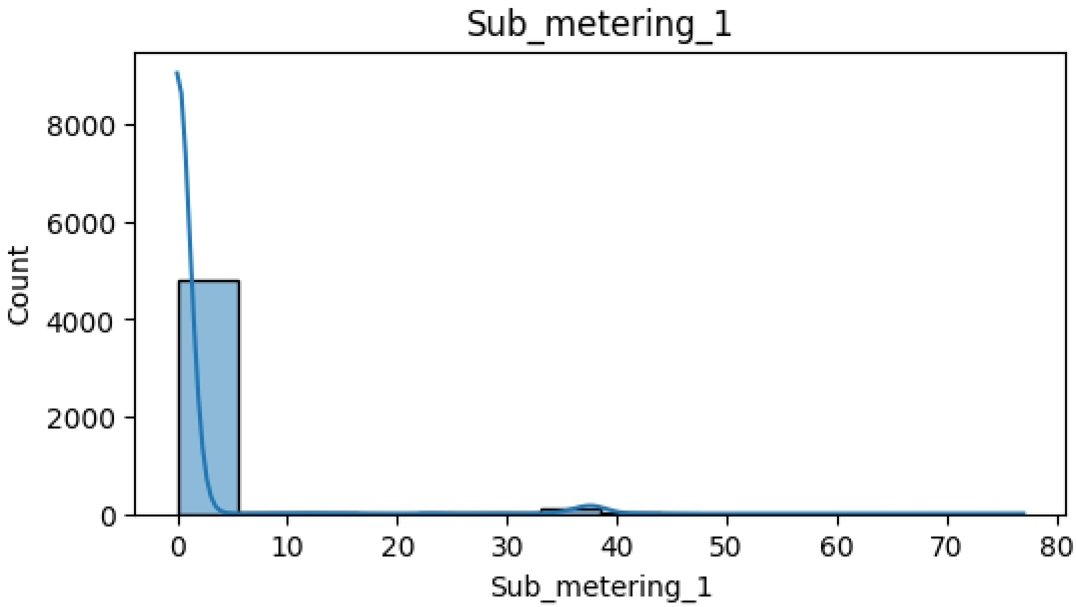
Structure of the data

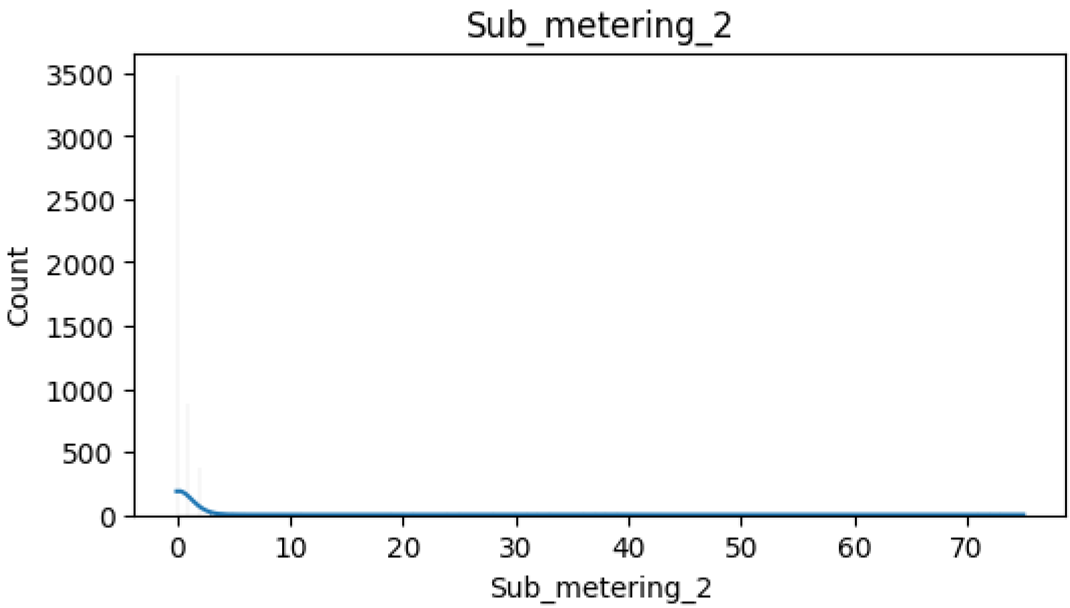
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2075259 entries, 0 to 2075258
Data columns (total 8 columns):
 #   Column                 Dtype
---  ------                 -----
 0   Datetime               datetime64[ns]
 1   Global_active_power    float64
 2   Global_reactive_power  float64
 3   Voltage                float64
 4   Global_intensity       float64
 5   Sub_metering_1         float64
 6   Sub_metering_2         float64
 7   Sub_metering_3         float64
dtypes: datetime64[ns](1), float64(7)
memory usage: 126.7 MB
```

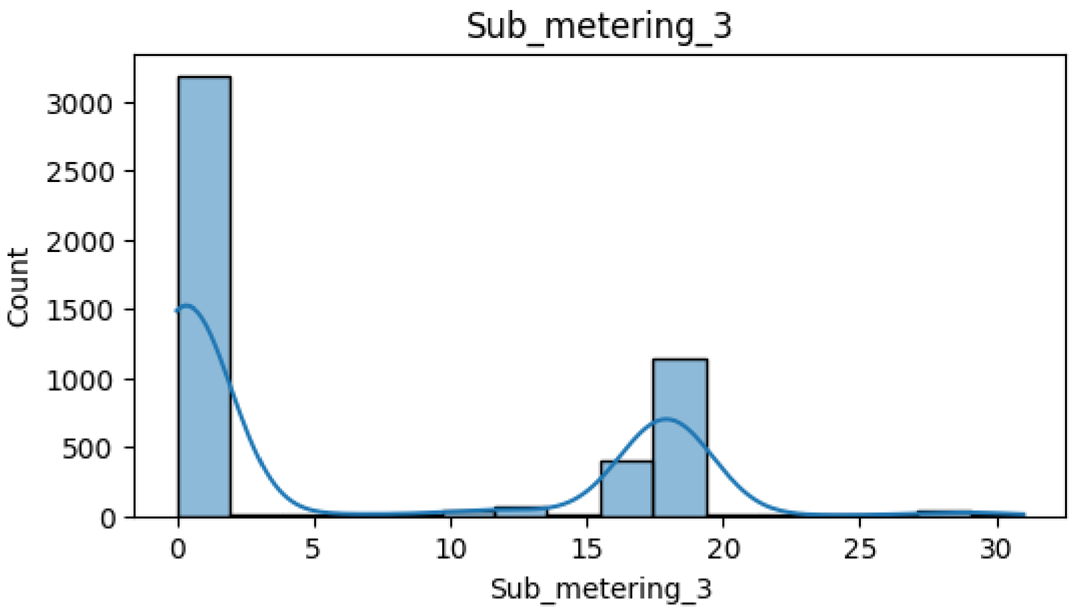
Data Distribution of each feature:




Global_intensity
Count
1000
800
600
400
200
0
0
10
20
30
40
Global_intensity


Sub_metering_1
Count
8000
6000
4000
2000
0
0
10
20
30
40
50
60
70
80
Sub_metering_1


Sub_metering_2
Count
3500
3000
2500
2000
1500
1000
500
0
0
10
20
30
40
50
60
70
Sub_metering_2

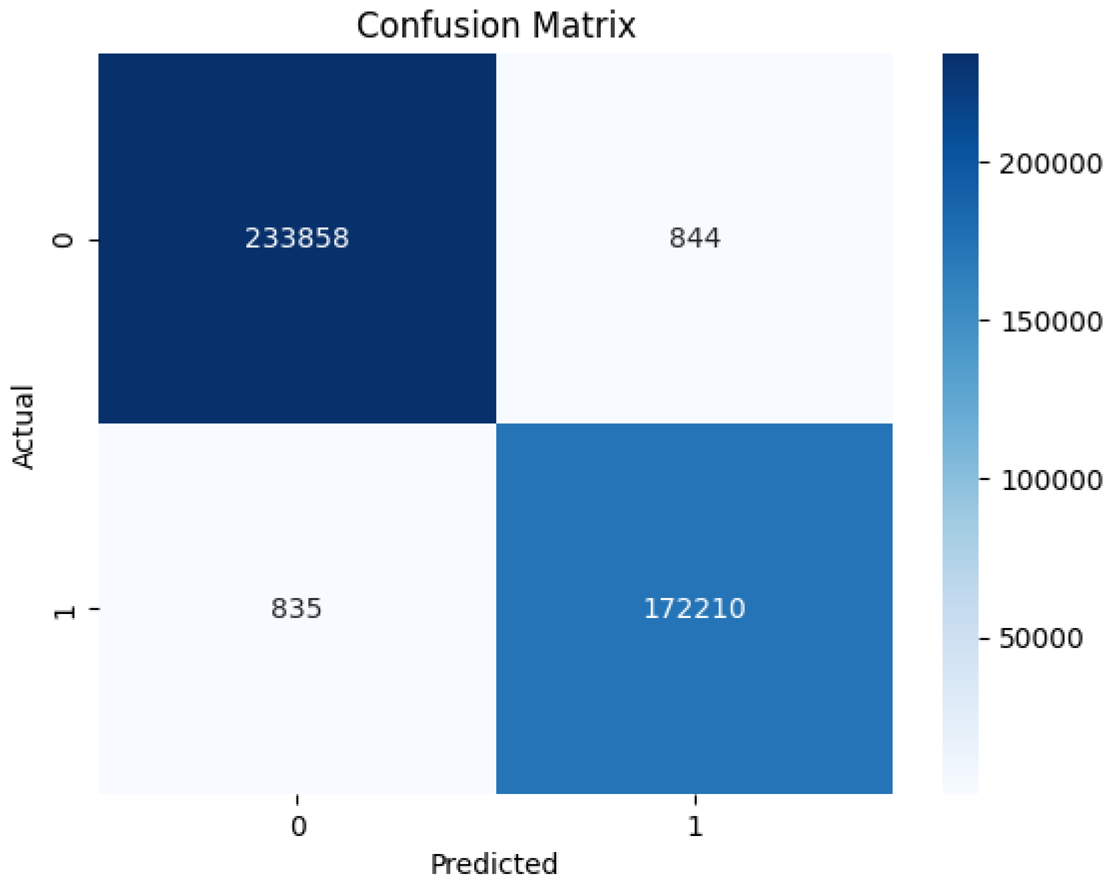

Sub_metering_3
Count
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
25
30
Sub_metering_3

Training report

```
Accuracy: 0.9958822505131859
[[233858    844]
 [   835 172210]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    234702
           1       1.00      1.00      1.00    173045

    accuracy                           1.00    407747
   macro avg       1.00      1.00      1.00    407747
weighted avg       1.00      1.00      1.00    407747
```



Project workflow and my approach:

During this week, a complete Machine Learning workflow was implemented using the Household Power Consumption dataset. The work focused on both regression and classification approaches for analyzing and predicting household energy consumption patterns.

The following tasks were completed:

- Defined the business problem and identified the prediction objectives.
- Collected and imported the Household Power Consumption dataset.
- Performed data preprocessing and cleaning.
- Conducted Exploratory Data Analysis (EDA).
- Converted data types and handled missing values.
- Created additional time-based features through feature engineering.
- Performed duplicate and outlier analysis.
- Conducted correlation analysis for feature selection.
- Developed and trained a Linear Regression model for energy consumption prediction.
- Evaluated regression model performance using standard metrics.
- Serialized and saved the trained regression model using Pickle.
- Created a classification target variable for energy usage categorization.
- Developed and trained a Logistic Regression model.
- Evaluated classification performance using accuracy, confusion matrix, and classification report.
- Visualized classification results.
- Serialized and saved the trained classification model.

Workflow Implementation

Step 1: Problem Definition

The objective of the project was to analyze household electrical consumption data and develop machine learning models capable of:

- Predicting total active power consumption.
- Classifying energy usage into different consumption categories.

The target variable selected was:

Global_active_power

Input features included:

- Voltage

- Global_intensity
- Global_reactive_power
- Sub_metering_1
- Sub_metering_2
- Sub_metering_3

Step 2: Data Collection and Import

The Household Power Consumption dataset was collected and imported into Python using Pandas.

Dataset Characteristics:

- Time-series household energy consumption data.
- Contains electrical measurements recorded over time.
- Data initially stored in text format with semicolon-separated values.

Activities performed:

- Imported dataset into a DataFrame.
- Parsed date and time columns.
- Combined date and time into a single datetime field.
- Loaded data into a structured format for further analysis.

Step 3: Data Preprocessing

Several preprocessing operations were performed to improve data quality.

Data Type Conversion

Numerical columns initially stored as object datatype were converted into numeric format.

Missing Value Handling

- Missing values were identified.
- Null records were removed due to the large dataset size.
- Data integrity was maintained while minimizing information loss.

Duplicate Handling

- Duplicate records were detected.
- Duplicate entries were removed.

Step 4: Feature Engineering

Additional features were extracted from the datetime column to capture temporal patterns.

Generated Features:

- Hour
- Day
- Month
- Weekday

These features help models understand energy consumption behavior across different time periods.

Step 5: Exploratory Data Analysis (EDA)

EDA was performed to understand data distribution and identify patterns.

Analysis conducted:

- Distribution plots
- Histogram analysis
- Boxplot analysis
- Correlation analysis

Key Observations

- Global Active Power showed a right-skewed distribution.
- Voltage exhibited relatively stable behavior.
- Sub-metering variables contained a large number of zero values.
- Energy consumption features demonstrated strong relationships with the target variable.

Step 6: Outlier Analysis

Outliers were examined using boxplots.

Observation:

High-value observations represented genuine energy consumption events rather than noise.

Decision:

- Outliers were retained.
- No outlier removal was performed to preserve real-world consumption behavior.

Step 7: Feature Selection

Correlation analysis was performed to identify the most relevant predictors.

Selected Features:

- Global_intensity

- Sub_metering_1
- Sub_metering_2
- Sub_metering_3
- Voltage
- Global_reactive_power

These features were used as input variables for model training.

Linear Regression Model Workflow

Data Splitting

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

Purpose:

- Train the model on historical data.
- Evaluate performance on unseen data.

Model Training

Algorithm Used:

Linear Regression

The model was trained to predict:

Global_active_power

using the selected electrical measurement features.

Model Evaluation

Evaluation Metrics:

- Mean Absolute Error (MAE)
- R^2 Score

Regression Output

[Insert Regression Metrics Screenshot Here]

Model Serialization

The trained model was saved using Pickle.

Generated File:

linear_regression_model.pkl

Purpose:

- Model reuse
- Deployment readiness
- Faster inference without retraining

Logistic Regression Model Workflow

Classification Problem Creation

To enable classification, a new target variable was generated.

Method:

- Mean value of Global Active Power was calculated.
- Values above the mean were labeled as High Usage (1).
- Values below the mean were labeled as Low Usage (0).

Generated Target:

usage_type

Data Splitting

The classification dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

Model Training

Algorithm Used:

Logistic Regression

Objective:

Predict whether household energy consumption belongs to:

- High Usage Category
- Low Usage Category

Model Evaluation

Evaluation Techniques:

- Accuracy Score
- Confusion Matrix
- Classification Report

Metrics evaluated:

- Accuracy
- Precision
- Recall
- F1-Score

Classification Output

[Insert Classification Metrics Screenshot Here]

Confusion Matrix

[Insert Confusion Matrix Screenshot Here]

Model Serialization

The trained classification model was saved using Pickle.

Generated File:

logistic_model.pkl

Overall Outcome

This week's implementation successfully established a complete end-to-end machine learning pipeline for household energy consumption analysis. The workflow included data collection, preprocessing, feature engineering, exploratory analysis, regression modeling, classification modeling, evaluation, and model persistence. Both Linear Regression and Logistic Regression models were developed and prepared for future integration into analytical or deployment environments.